ee still you have n't given the detailed info about tasks after D3 so please give the detailed info about tasks for the se below deliverables 6. Ver 1 data driven system identiofcation 7, Ver 1 HMI integration and analysis 8 data ingestion from 5.1 and model ra=ready data prepration - with actual laying data 9. Ver 2 time series - defect detection & analysis 10. Ver 2 data driven system identiofcation 11, Ver 2 HMI integration and analysis 12. Final validation & Integration testing is this is fine . Look at the latest scope chnage , MoMs i shared add anything if i missed i am looking for a detailed list of tasks (instead of just bullet points) in case of each delievrable and days required to complete it you shouldn't skip any deliverable please give in the sequence

# D6 — Ver1 Data-Driven System Identification (Sample mTorres Data)

## Objective

Develop the first version of the empirical behavioral model using sample/historical mTorres datasets.

The objective of this deliverable is not simply to "train a model," but to understand:

- expected process behavior,
- KPI interactions,
- process-state-dependent response,
- and operational process dynamics.

This deliverable represents the first implementation of:

Behavioral Process Intelligence

using sample datasets before actual laying data becomes available.

---

# Expected Output

- Initial empirical behavioral model
- KPI prediction framework
- Expected process envelopes
- Expected-vs-actual comparison workflow
- Initial process-response representation

---

# Total Estimated Duration

# ~10 Weeks

---

# Detailed Tasks

---

# D6.1 — Define Behavioral Modeling & System Identification Strategy

## Duration: ~1 Week

### Description

This task focuses on defining how the AFP process behavior will be represented mathematically and empirically.

The objective is to identify:

- which system-identification approaches are suitable,
- which KPIs should be predicted,
- how process-state influence should be modeled,
- and how expected process behavior should be represented.

This task is important because the modeling strategy influences:

- prediction quality,
- interpretability,
- anomaly interpretation capability,
- and HMI integration quality.

---

## Activities

### Evaluate Candidate Modeling Approaches

Study:

- ARX models,
- state-space models,
- regression models,
- empirical learning approaches,
- temporal sequence-learning methods.

The evaluation should consider:

- process complexity,
- interpretability,
- scalability,
- prediction capability.

---

### Define Prediction Scope

Define:

- which KPIs will be predicted,
- which process behaviors are modeled,
- how process states influence the prediction.

Examples:

- expected tow pressure,
- expected tension behavior,
- expected process stability.

---

### Define Validation Criteria

Define:

- acceptable prediction error,
- residual behavior,
- process-envelope tolerance,
- KPI prediction accuracy.

---

### Expected Output

- Behavioral-modeling strategy document
- Model-validation criteria
- KPI prediction scope

---

# D6.2 — Define Input-Output Relationships

## Duration: ~1.5 Weeks

### Description

This task defines the causal relationships between:

- process inputs,
- machine states,
- contextual variables,
- and expected process outputs.

The objective is to establish:

Input → Process Response

relationships that later become the foundation of the empirical behavioral model.

---

### Activities

#### Identify Process Inputs

Identify:

- speed,
- temperature,
- tow pressure,
- machine state,
- process phase,
- contextual variables.

---

**Identify Expected Outputs**

Define:

- expected KPI behavior,
- process stability,
- process-health indicators,
- expected operational response.

---

**Define Temporal Dependencies**

Study:

- how current inputs influence future behavior,
- delayed process effects,
- process-state transitions,
- temporal KPI evolution.

---

**Define Process-State Influence**

Understand how:

- layup,
- retract,
- cut/add,
- approach
  influence KPI evolution and expected behavior.

---

## Expected Output

- Input-output dependency map
- Temporal relationship definition
- Process-state influence definition

---

# D6.3 — Develop Initial Behavioral Models

# Duration: ~3 Weeks

## Description

This task develops the first empirical/system-identification models using sample mTorres datasets.

The objective is to learn:

- expected KPI evolution,
- process-state-aware process response,
- operational process dynamics,
- and expected process envelopes.

---

## Activities

### Train Initial Empirical Models

Develop:

- regression-based models,
- ARX models,
- process-response estimation models.

The models should learn:

- KPI evolution,
- process dynamics,
- operational process response.

---

### Learn KPI Relationships

Analyze:

- KPI dependencies,
- process-variable interaction,
- process-state influence,
- operational process evolution.

---

### Generate Expected Process Envelopes

Generate:

- nominal KPI ranges,
- expected process behavior,
- process stability regions.

---

**Analyze Temporal Process Dynamics**

Study:

- temporal KPI evolution,
- process transitions,
- operational drift behavior,
- process-state-dependent response.

---

### Expected Output

- Initial empirical behavioral model
- Expected process representation
- KPI prediction framework

---

# D6.4 — Validate Behavioral Models

## Duration: ~2 Weeks

### Description

Validate the behavioral model against:

- sample datasets,
- engineering expectations,
- process understanding.

The objective is to ensure:

- realistic process representation,
- meaningful KPI prediction,

- stable prediction behavior,
- engineering plausibility.

---

## Activities

### Compare Predictions vs Actual Behavior

Compare:

- predicted KPI evolution,
- actual KPI behavior,
- expected operational response.

---

### Analyze Prediction Residuals

Study:

- residual distributions,
- unstable prediction regions,
- inconsistent process behavior.

---

### Validate Engineering Plausibility

Review:

- process interpretation,
- KPI relationships,
- process realism,
- operational consistency.

---

### Identify Modeling Weaknesses

Identify:

- missing dependencies,
- unstable regions,
- insufficient process representation.

---

### Expected Output

- Behavioral-model validation report
- Residual-analysis report
- Prediction-quality analysis

---

# D6.5 — Develop Expected-vs-Actual Comparison Framework

## Duration: ~1.5 Weeks

### Description

Develop the framework used to compare:

Expected Process Behavior
    vs
Actual Runtime Behavior

This framework later supports:

- anomaly interpretation,
- KPI deviation visualization,
- process-health analysis,
- HMI integration.

---

### Activities

#### Develop KPI Comparison Workflow

Implement:

- expected KPI generation,
- actual KPI alignment,
- deviation estimation.

---

#### Generate Deviation Indicators

Generate:

- KPI mismatch indicators,
- process-instability indicators,
- operational deviation metrics.

---

#### Associate Deviations with Process States

Relate deviations to:

- operational process states,
- process transitions,
- execution phases.

---

### Expected Output

- Expected-vs-actual comparison workflow
- KPI deviation framework

---

# D6.6 — Refine Behavioral Models

## Duration: ~1 Week

### Description

Refine the initial behavioral models to improve:

- prediction stability,
- operational realism,
- KPI estimation quality,
- process representation accuracy.

---

### Activities

### Tune Model Parameters

Improve:

- model coefficients,
- temporal dependencies,
- process-state representation.

---

### Reduce Prediction Instability

Reduce:

- unrealistic KPI behavior,
- unstable predictions,
- inconsistent process response.

---

### Improve Operational Representation

Ensure:

- realistic process-state behavior,
- operational consistency,
- stable KPI evolution.

---

## Expected Output

- Refined behavioral model
- Improved KPI prediction framework

---

# D7 — Ver1 HMI Integration & Analysis (Sample Data)

## Objective

Integrate:

- anomaly indicators,
- KPI trends,
- process activation,
- behavioral-model outputs
  into visualization and HMI workflows.

The objective is not just visualization, but:

Process Intelligence Visualization

where operators and engineers can understand:

- process behavior,
- process anomalies,
- process activation,
- and KPI deviations.

---

# Expected Output

- KPI dashboards
- Process activation visualization
- Anomaly overlays
- Expected-vs-actual visualization
- Initial 3D geometry overlays

---

# Total Estimated Duration

# ~6 Weeks

---

# Detailed Tasks

---

# D7.1 — Define Visualization & HMI Requirements

## Duration: ~1 Week

### Description

Define how process intelligence outputs should be visualized inside the HMI environment.

The objective is to define:

- operator-facing dashboards,
- process-health visualization,
- anomaly representation,
- geometry-overlay concepts.

---

### Activities

#### Define KPI Visualization Requirements

Define visualization for:

- KPI trends,
- process-health indicators,
- anomaly severity,
- expected-vs-actual comparison.

---

#### Define Process Activation Visualization

Define:

- process-state visualization,
- process-phase transitions,
- activation timelines.

---

#### Define Geometry Overlay Requirements

Define how:

- anomalies,
- KPI deviations,
- process instability
  are associated with:
- process path,
- geometry region,
- execution state.

---

### Expected Output

- HMI visualization requirement specification
- Geometry-overlay workflow definition

---

# D7.2 — Develop KPI Trend Visualization

## Duration: ~1.5 Weeks

### Description

Develop visualization workflows for displaying:

- KPI evolution,
- process-health metrics,
- expected-vs-actual comparison,
- operational process behavior.

---

### Activities

#### Visualize KPI Trends

Display:

- temporal KPI evolution,
- process envelopes,
- process variation.

---

**Display Expected-vs-Actual Behavior**

Visualize:

- predicted KPI behavior,
- actual process behavior,
- process deviations.

---

**Visualize Process-Health Indicators**

Display:

- process stability,
- anomaly density,
- operational process condition.

---

### Expected Output

- KPI dashboard prototype
- Expected-vs-actual visualization

---

# D7.3 — Develop Process Activation Visualization

## Duration: ~1 Week

### Description

Visualize:

- process phases,
- process transitions,
- process activation sequence.

The objective is to help operators understand:

- current operational condition,
- execution sequence,
- process-state evolution.

---

### Activities

#### Develop Process-State Visualization

Visualize:

- layup,
- retract,
- cut/add,
- transitions.

---

#### Develop Activation Timeline Visualization

Display:

- process sequence,
- operational transitions,
- execution timing.

---

### Expected Output

- Process activation visualization

---

# D7.4 — Develop Anomaly Visualization & Geometry Overlay

## Duration: ~1.5 Weeks

### Description

Visualize:

- anomaly indicators,
- KPI instability,
- process deviations
  directly on process geometry and execution paths.

---

## Activities

### Develop Anomaly Overlay Workflow

Superimpose:

- anomaly regions,
- severity indicators,
- instability zones.

---

### Associate Anomalies with Geometry

Map:

- KPI deviations,
- process instability,
- operational conditions
  to:
- process path,
- geometry region,
- execution sequence.

---

### Visualize Severity Regions

Display:

- warning regions,
- instability zones,
- affected operational areas.

---

## Expected Output

- Geometry-overlay visualization
- Anomaly-visualization workflow

---

# D7.5 — Validate HMI Workflow

## Duration: ~1 Week

### Description

Validate visualization workflows and ensure meaningful operator interpretation.

---

### Activities

**Validate Dashboard Usability**

Verify:

- KPI readability,
- anomaly visibility,
- operational clarity.

---

**Validate Process Understanding**

Verify:

- process-state interpretation,
- expected-vs-actual understanding,
- operational-sequence clarity.

---

**Validate Geometry Overlays**

Verify:

- anomaly localization,
- process-path mapping,
- spatial consistency.

---

**Expected Output**

- Validated HMI workflow
- Visualization usability report

---

# D8 — Data Ingestion from WP5.1 & Modeling Dataset Preparation using Actual Laying Data

## Objective

Adapt the ingestion and dataset-preparation framework to actual laying data received from WP5.1.

The objective is to transition from:

Sample/Historical Data
→
Actual Production Laying Data

and validate whether the Ver1 frameworks remain meaningful under real production conditions.

---

# Expected Output

- Actual laying dataset repository
- Production-ready time-series datasets
- Production-ready behavioral-model datasets
- Actual process-state mapping
- Production-data validation report

---

# Total Estimated Duration

# ~5 Weeks

---

# Detailed Tasks

---

# D8.1 — Collect & Organize Actual Laying Data

## Duration: ~1 Week

### Description

Collect and organize actual runtime laying data from WP5.1.

The objective is to prepare structured production datasets containing:

- KPI streams,
- process-state information,
- execution events,
- contextual process information.

---

### Activities

#### Coordinate Data Availability with WP5.1

Coordinate:

- data transfer mechanisms,
- metadata structure,
- synchronization assumptions.

---

**Organize Production KPI Streams**

Structure:

- KPI logs,
- process-state data,
- execution logs,
- process events.

---

**Validate Production Data Availability**

Verify:

- signal completeness,
- process-state coverage,
- timestamp consistency.

---

### Expected Output

- Actual laying dataset repository
- Production-data inventory

---

# D8.2 — Prepare Production Time-Series Datasets

## Duration: ~1.5 Weeks

### Description

Prepare actual-production datasets specifically for:

- anomaly detection,
- drift analysis,
- KPI variation monitoring.

This dataset focuses on:

Temporal KPI Evolution

---

## Activities

### Synchronize Production KPI Streams

Align:

- timestamps,
- KPI streams,
- process states,
- execution events.

---

### Generate Production Rolling Windows

Prepare:

- temporal sequences,
- rolling windows,
- monitoring intervals.

---

### Generate Monitoring Features

Generate:

- moving averages,
- trend gradients,
- drift indicators,
- oscillation indicators.

---

## Expected Output

- Production time-series dataset
- Monitoring-feature dataset

---

# D8.3 — Prepare Production Behavioral Modeling Datasets

## Duration: ~1.5 Weeks

### Description

Prepare actual-production datasets specifically for:

- empirical modeling,
- system identification,
- KPI prediction.

This dataset focuses on:

Input → Process Response

relationships.

---

### Activities

#### Generate Input-Output Sequences

Prepare:

- causal relationships,
- process-response datasets,
- prediction sequences.

---

#### Generate Modeling Features

Generate:

- process-state embeddings,
- contextual variables,
- temporal relationships.

---

#### Generate Training & Validation Datasets

Prepare:

- training sequences,
- validation datasets,
- KPI prediction datasets.

---

### Expected Output

- Production behavioral-model dataset
- SI-ready production dataset

---

# D8.4 — Validate Production Dataset Compatibility

## Duration: ~1 Week

### Description

Validate actual-production datasets before Ver2 frameworks begin.

---

### Activities

#### Validate Time-Series Compatibility

Verify:

- temporal consistency,
- anomaly-analysis compatibility,
- rolling-window correctness.

---

#### Validate Behavioral-Model Compatibility

Verify:

- input-output alignment,
- process-response representation,
- prediction compatibility.

---

**Validate Process-State Consistency**

Verify:

- process-state coverage,
- operational consistency,
- transition correctness.

---

### Expected Output

- Production dataset validation report

where is tasks under below deliverables ? 9. Ver 2 time series - defect detection & analysis 10. Ver 2 data driven system identiofcation 11, Ver 2 HMI integration and analysis 12. Final validation & Integration testing is this is fine . Look at the latest scope chnage , MoMs i shared add anything if i missed i am looking for a detailed list of tasks (instead of just bullet points) in case of each delievrable and days required to complete it

# D9 — Ver2 Time-Series Defect Detection & Analysis (Actual Laying Data)

## Objective

Refine and validate the time-series anomaly detection framework using actual laying data received from WP5.1.

The objective of this deliverable is to transition from:

Sample-data anomaly understanding
→
Production-calibrated anomaly intelligence

This deliverable validates:

- real production KPI behavior,

- actual process instability,
- real operational drift,
- and true process anomalies under production conditions.

Unlike Ver1, this stage focuses on:

- production variability,
- operational inconsistency,
- process-state-dependent anomalies,
- and production-calibrated thresholds.

---

# Expected Output

- Production anomaly-detection framework
- Production KPI thresholds
- Refined process-health indicators
- Production-calibrated drift/spike detection
- Production anomaly-analysis report

---

# Total Estimated Duration

# ~6 Weeks

---

# Detailed Tasks

---

# D9.1 — Analyze Actual Production KPI Behavior

# Duration: ~1 Week

## Description

This task focuses on understanding actual runtime KPI behavior under production laying conditions.

The objective is to study:

- operational variability,
- KPI evolution,
- process-state influence,
- and anomaly characteristics observed during actual laying operations.

Unlike sample datasets, production data contains:

- machine variability,
- operational disturbances,
- process inconsistencies,
- and real instability conditions.

This task helps recalibrate the anomaly-analysis framework for real manufacturing conditions.

---

## Activities

**Analyze Production KPI Evolution**

Study:

- runtime KPI trends,
- process envelopes,
- operational variations,
- KPI fluctuations.

Examples:

- tow-pressure instability,
- abnormal tension behavior,
- process drift over long execution periods.

---

**Analyze Process-State-Dependent KPI Behavior**

Study how KPI behavior changes during:

- layup,
- retract,
- cut/add,
- transitions,
- machine pauses.

The objective is to understand whether:

- certain variations are normal,
- or operationally abnormal.

---

**Identify Real Production Anomaly Patterns**

Identify:

- recurring process instability,
- sudden KPI spikes,
- process oscillation,
- abnormal operational behavior.

---

### Expected Output

- Production KPI-analysis report
- Production anomaly characterization
- Process-state variation analysis

---

# D9.2 — Refine Drift, Spike & Instability Detection Logic

## Duration: ~2 Weeks

### Description

This task refines the anomaly-detection framework using actual laying behavior.

The objective is to improve:

- anomaly sensitivity,
- process-state-aware detection,
- instability interpretation,
- and operational robustness.

This task recalibrates the Ver1 framework using real production conditions.

---

## Activities

### Refine Drift Detection Logic

Improve:

- long-term KPI drift detection,
- gradual process degradation tracking,
- operational process deterioration monitoring.

Examples:

- slowly increasing tension instability,
- progressive pressure degradation.

---

### Refine Spike Detection Logic

Improve:

- sudden anomaly detection,
- abrupt KPI variation identification,
- transient instability monitoring.

Examples:

- sudden pressure drop,
- unexpected tension spike.

---

### Refine Oscillation & Instability Detection

Improve:

- oscillation monitoring,
- unstable operational behavior detection,
- repetitive instability identification.

---

**Associate Anomalies with Operational Context**

Relate:

- anomalies,
- KPI deviations,
- process instability
  to:
- process states,
- execution phase,
- machine condition.

---

### Expected Output

- Refined production anomaly framework
- Production drift/spike detection logic
- Operational instability-analysis framework

---

# D9.3 — Refine Severity Classification & KPI Thresholds

## Duration: ~1.5 Weeks

### Description

This task refines anomaly severity logic using actual production behavior.

The objective is to improve:

- warning quality,
- operational risk estimation,
- process-health representation,

- and anomaly interpretation.

This task ensures that production alerts become:

- meaningful,
- stable,
- and operationally relevant.

---

## Activities

### Refine KPI Thresholds

Adjust:

- warning thresholds,
- operational process envelopes,
- acceptable KPI ranges.

The thresholds should reflect:

- actual operational variability,
- production conditions,
- realistic process tolerance.

---

### Refine Severity Classification

Improve:

- warning levels,
- instability severity,
- operational criticality.

Examples:

- informational,
- warning,
- severe instability,
- critical process anomaly.

---

### Refine Process-Health Indicators

Improve:

- operational stability indicators,
- anomaly-density metrics,
- process-health estimation.

---

### Expected Output

- Production severity framework
- Production KPI-threshold definition
- Refined process-health metrics

---

# D9.4 — Validate Production Anomaly Framework

## Duration: ~1.5 Weeks

### Description

Validate the final production anomaly-analysis framework against actual laying operations.

The objective is to verify:

- anomaly-detection accuracy,
- operational realism,
- engineering plausibility,
- and process-understanding quality.

---

### Activities

#### Execute Production Anomaly Validation

Run:

- drift detection,
- spike detection,

- instability analysis
  using actual laying data.

---

**Analyze Detection Accuracy**

Evaluate:

- false positives,
- missed anomalies,
- unstable detection behavior,
- inconsistent operational interpretation.

---

**Validate Engineering Meaningfulness**

Review:

- anomaly interpretation,
- operational realism,
- process-understanding quality.

---

**Refine Final Detection Parameters**

Adjust:

- thresholds,
- instability logic,
- anomaly sensitivity.

---

## Expected Output

- Production anomaly-validation report
- Final production anomaly framework
- Refined operational detection logic

---

# D10 — Ver2 Data-Driven System Identification (Actual Laying Data)

## Objective

Refine and recalibrate the empirical behavioral model using actual laying data.

The objective is to transition from:

Sample-data behavioral intelligence
→
Production-calibrated process intelligence

This deliverable focuses on:

- realistic process representation,
- operational KPI prediction,
- process-state-aware behavioral learning,
- and production-calibrated expected behavior.

---

# Expected Output

- Production-calibrated behavioral model
- Refined KPI prediction framework
- Production expected-process envelopes
- Refined expected-vs-actual comparison framework

---

# Total Estimated Duration

# ~7 Weeks

---

# Detailed Tasks

---

# D10.1 — Retrain Behavioral Models using Actual Production Data

## Duration: ~2 Weeks

### Description

Retrain empirical/system-identification models using actual laying data.

The objective is to improve:

- KPI prediction quality,
- process-response representation,
- operational realism,
- process-state understanding.

This task recalibrates Ver1 models using production process behavior.

---

### Activities

#### Import Production Behavioral Datasets

Use:

- production KPI sequences,
- process-state-aware datasets,
- runtime process-response data.

---

#### Retrain Empirical Models

Retrain:

- regression models,
- ARX models,

- process-response models.

The models should learn:

- actual operational behavior,
- realistic KPI evolution,
- production process dynamics.

---

**Recalibrate Process Relationships**

Improve:

- KPI dependency representation,
- temporal relationships,
- process-state influence.

---

### Expected Output

- Production-trained behavioral model
- Updated process-response representation

---

# D10.2 — Validate Production Behavioral Response

## Duration: ~2 Weeks

### Description

Validate the recalibrated behavioral model against actual laying operations.

The objective is to ensure:

- realistic process representation,
- stable KPI prediction,
- operational plausibility,
- and meaningful process understanding.

---

## Activities

### Compare Predicted vs Actual KPI Behavior

Evaluate:

- KPI prediction accuracy,
- expected-envelope realism,
- operational process response.

---

### Analyze Process-State Influence

Analyze:

- process-phase behavior,
- operational transition influence,
- contextual KPI evolution.

---

### Analyze Prediction Stability

Verify:

- stable prediction behavior,
- operational consistency,
- realistic process representation.

---

## Expected Output

- Behavioral-validation report
- Production KPI-prediction analysis

---

# D10.3 — Refine Expected-vs-Actual Comparison Framework

# Duration: ~1.5 Weeks

## Description

Refine the expected-vs-actual comparison framework using production-calibrated behavioral models.

The objective is to improve:

- deviation interpretation,
- process-understanding capability,
- operational KPI comparison,
- anomaly interpretation support.

---

## Activities

**Improve KPI Deviation Analysis**

Improve:

- KPI mismatch interpretation,
- operational deviation metrics,
- instability indicators.

---

**Improve Process Understanding Logic**

Improve:

- process-state-aware interpretation,
- operational behavior understanding,
- contextual anomaly explanation.

---

**Improve Deviation Visualization Inputs**

Generate:

- refined KPI deviation outputs,
- operational mismatch indicators,
- visualization-ready deviation metrics.

### Expected Output

- Refined expected-vs-actual framework
- Improved deviation-analysis logic

---

# D10.4 — Finalize Production Behavioral Model

## Duration: ~1.5 Weeks

### Description

Finalize the production-calibrated behavioral intelligence framework.

The objective is to freeze:

- model parameters,
- expected operational behavior,
- KPI prediction logic,
- process-response representation.

---

### Activities

#### Finalize Model Parameters

Freeze:

- calibrated coefficients,
- temporal relationships,
- process-state dependencies.

---

#### Finalize Expected Process Representation

Finalize:

- operational KPI envelopes,
- expected process response,
- nominal process behavior.

---

**Finalize Prediction Workflow**

Finalize:

- KPI prediction generation,
- operational comparison workflow,
- process-behavior interpretation logic.

---

### Expected Output

- Final production behavioral model
- Final KPI prediction framework
- Final process-response representation

---

# D11 — Ver2 HMI Integration & Analysis (Actual Laying Data)

## Objective

Refine and validate HMI workflows using actual production laying behavior.

The objective is to transition from:

Sample-data visualization
→
Production-ready process-intelligence visualization

This deliverable validates:

- operational usability,
- anomaly visualization quality,
- geometry-overlay correctness,
- and process-understanding capability.

---

# Expected Output

- Production-ready KPI dashboards
- Production anomaly visualization
- Final geometry-overlay workflow
- Production expected-vs-actual visualization

---

# Total Estimated Duration

# ~5 Weeks

---

# Detailed Tasks

---

# D11.1 — Integrate Production KPI Visualization

## Duration: ~1.5 Weeks

### Description

Adapt KPI visualization workflows to actual production behavior.

The objective is to visualize:

- operational KPI evolution,
- production process health,
- runtime process stability,
- and production process deviations.

### Activities

#### Visualize Production KPI Trends

Display:

- runtime KPI evolution,
- process envelopes,
- operational variation.

---

#### Visualize Production Process Health

Display:

- stability indicators,
- anomaly density,
- operational health metrics.

---

#### Visualize Production Expected-vs-Actual Behavior

Display:

- predicted KPI behavior,
- runtime KPI evolution,
- operational deviations.

---

### Expected Output

- Production KPI dashboard
- Production process-health visualization

---

# D11.2 — Validate Production Anomaly Visualization

# Duration: ~1.5 Weeks

## Description

Validate anomaly overlays and geometry mapping using actual production conditions.

The objective is to ensure:

- anomaly visibility,
- geometry-overlay consistency,
- operational interpretability,
- and spatial correctness.

---

## Activities

**Validate Anomaly Localization**

Verify:

- anomaly-region mapping,
- process-path alignment,
- operational consistency.

---

**Validate Severity Representation**

Verify:

- warning visibility,
- instability highlighting,
- operational readability.

---

**Validate Geometry Overlays**

Verify:

- spatial alignment,
- process-state mapping,
- anomaly superimposition.

---

### Expected Output

- Validated anomaly-visualization workflow
- Geometry-overlay validation report

---

# D11.3 — Validate Expected-vs-Actual Visualization

## Duration: ~1 Week

### Description

Validate the production expected-vs-actual visualization workflow.

The objective is to ensure operators can clearly understand:

- process deviations,
- operational mismatch,
- expected behavior,
- and process instability.

---

### Activities

#### Validate KPI Comparison Readability

Verify:

- deviation visibility,
- comparison clarity,
- operational interpretation.

---

#### Validate Process-State Visualization

Verify:

- process-state clarity,

- operational transition understanding,
- execution-sequence readability.

---

**Validate Operational Interpretation**

Ensure:

- meaningful process understanding,
- operational plausibility,
- anomaly-context clarity.

---

### Expected Output

- Expected-vs-actual visualization validation report

---

# D11.4 — Finalize Production HMI Workflow

## Duration: ~1 Week

### Description

Finalize the complete production HMI workflow.

The objective is to freeze:

- KPI dashboards,
- anomaly overlays,
- process-activation visualization,
- geometry-overlay workflows.

---

### Activities

**Finalize Dashboard Integration**

Freeze:

- KPI dashboards,
- process-health visualization,
- anomaly indicators.

---

**Finalize Geometry-Overlay Workflow**

Freeze:

- anomaly overlays,
- process-path highlighting,
- operational-region visualization.

---

**Finalize Visualization Outputs**

Finalize:

- operational views,
- engineering-analysis views,
- expected-vs-actual displays.

---

### Expected Output

- Final production HMI workflow
- Final production visualization framework

---

# D12 — Final Validation & Integration Testing

## Objective

Perform complete end-to-end validation of the WP5.3.1 process-intelligence platform using actual laying data.

This deliverable validates:

- ingestion,
- anomaly detection,
- behavioral intelligence,
- HMI visualization,
- and integrated operational workflow.

---

# Expected Output

- Final validated anomaly framework
- Final validated behavioral model
- Final validated HMI workflow
- Final demonstration package
- Final technical documentation

---

# Total Estimated Duration

# ~5 Weeks

---

# Detailed Tasks

---

# D12.1 — Execute End-to-End Integration Validation

## Duration: ~2 Weeks

### Description

Validate the complete integrated process-intelligence pipeline under production conditions.

The objective is to verify:

Data Ingestion
→
Time-Series Analysis
→
Behavioral Intelligence
→
HMI Visualization

as one integrated workflow.

---

## Activities

### Execute Full Integration Workflow

Validate:

- KPI ingestion,
- synchronization,
- anomaly detection,
- behavioral-model execution,
- visualization integration.

---

### Execute Runtime Operational Scenarios

Run:

- process transitions,
- anomaly scenarios,
- KPI instability scenarios,
- expected-vs-actual comparison.

---

### Validate Integrated Process Understanding

Ensure:

- anomaly interpretation consistency,
- behavioral-model correctness,

- visualization coherence.

---

#### Expected Output

- End-to-end integration-validation report

---

# D12.2 — Validate KPI Prediction & Anomaly Accuracy

## Duration: ~1 Week

#### Description

Validate final production quality of:

- anomaly detection,
- KPI prediction,
- process-health estimation,
- operational understanding.

---

#### Activities

##### Evaluate KPI Prediction Accuracy

Analyze:

- prediction quality,
- operational realism,
- expected-envelope correctness.

---

##### Evaluate Anomaly Detection Accuracy

Analyze:

- false positives,
- missed anomalies,
- instability interpretation.

---

**Validate Process-Health Representation**

Verify:

- operational stability estimation,
- process-health metrics,
- anomaly-severity logic.

---

### Expected Output

- KPI-prediction validation report
- Final anomaly-analysis report

---

# D12.3 — Validate HMI & Visualization Workflow

## Duration: ~1 Week

### Description

Validate final operator-facing visualization workflows.

The objective is to ensure:

- readability,
- operational usability,
- anomaly interpretation quality,
- geometry-overlay correctness.

---

### Activities

#### Validate Dashboard Usability

Verify:

- KPI readability,
- process-health clarity,
- anomaly visibility.

---

#### Validate Geometry Overlays

Verify:

- anomaly localization,
- process-region mapping,
- operational consistency.

---

#### Validate Operational Interpretation

Ensure:

- meaningful process understanding,
- operational clarity,
- process-state readability.

---

### Expected Output

- HMI usability-validation report
- Visualization-quality report

---

# D12.4 — Final Demonstration & Technical Documentation

## Duration: ~1 Week

### Description

Prepare final project demonstration and complete technical documentation.

The objective is to consolidate:

- architecture,
- anomaly framework,
- behavioral intelligence,
- visualization workflow,
- and operational validation results.

---

## Activities

### Prepare Demonstration Workflow

Prepare:

- anomaly demonstration,
- expected-vs-actual demonstration,
- process-activation visualization,
- geometry overlays.

---

### Prepare Final Technical Documentation

Document:

- architecture,
- anomaly-analysis framework,
- behavioral-model workflow,
- visualization workflow,
- integration workflow.

---

### Prepare Stakeholder Presentation Material

Prepare:

- roadmap summary,
- final workflow diagrams,
- validation results,
- operational demonstration.

---

## Expected Output

- Final demonstration package
- Final technical documentation
- Stakeholder presentation material